

PATIALA, India

WMO: 421010

Lat: 30.3578N

Lon: 76.4499E

Elev: 251

StdP: 98.35

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	6.3	3.0	4.8	13.2	4.3	5.3	12.1	4.4	14.7	3.5	15.4	0.4	320	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.5	41.8	24.6	40.3	24.4	38.7	24.5	29.5	33.9	29.1	33.7	28.7	33.2	1.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	25.7	31.8	28.1	25.1	31.6	27.7	24.5	31.3	100.0	34.5	98.1	34.4	96.1	33.2	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.8	3.1	2.5	3.1	44.2	1.6	1.0	1.9	44.9	0.9	45.5	0.0	46.0	-1.2	46.8		
			3.0	30.1	1.7	0.6	1.8	30.6	0.8	30.9	-0.1	31.3	-1.4	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	12.5	16.1	21.3	27.7	31.5	32.0	30.3	29.5	28.3	25.1	19.8	14.6
	DBStd	7.08	2.34	2.44	3.08	2.98	2.67	2.80	1.90	1.59	1.67	2.22	2.18	2.44
	HDD10.0	7	6	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	386	180	69	8	0	0	0	0	0	0	0	11	118
	CDD10.0	5158	84	172	351	532	667	659	631	606	549	469	294	143
	CDD18.3	2495	0	8	101	282	409	409	372	348	299	211	55	1
	CDH23.3	32406	1	56	949	3659	6167	6221	5139	4494	3404	1899	406	11
	CDH26.7	17662	0	4	345	2173	3977	3941	2732	2121	1491	809	68	0
Wind	WSAvg	0.7	0.7	0.9	0.9	1.0	1.1	1.0	0.8	0.6	0.6	0.4	0.4	0.5
Precipitation	PrecAvg	599	18	26	17	16	22	87	154	144	93	9	2	10
	PrecMax	1181	68	75	73	77	171	460	535	490	273	90	20	59
	PrecMin	306	0	0	0	0	0	8	9	0	0	0	0	0
	PrecStd	211	20	21	20	18	34	92	118	119	84	22	5	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	27.2	34.6	41.1	43.7	44.0	38.0	35.7	34.7	34.0	29.9	24.8
		MCWB	16.0	19.6	22.5	23.7	25.3	25.4	28.2	28.1	27.1	24.5	19.7	16.6
	2%	DB	20.4	25.0	32.2	39.2	42.0	42.0	36.4	34.6	33.9	33.0	28.5	23.0
		MCWB	14.9	18.2	21.4	23.2	24.0	25.5	28.3	28.3	27.1	24.4	19.1	16.0
	5%	DB	19.1	23.3	30.2	37.7	40.6	40.4	35.4	33.8	33.2	32.0	27.2	21.5
		MCWB	14.1	17.2	21.0	22.5	23.8	25.5	28.3	28.2	27.1	24.0	18.6	15.5
	10%	DB	17.8	21.8	28.3	36.0	39.1	38.5	34.3	33.1	32.5	31.0	25.9	20.1
		MCWB	13.4	16.1	20.2	21.9	24.1	25.8	28.2	28.1	26.9	23.6	18.2	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	20.7	24.9	28.6	29.5	29.7	29.7	29.7	29.4	27.7	22.1	18.8
		MCDB	20.5	25.3	32.0	38.1	38.6	35.2	33.7	33.3	33.0	31.8	27.2	22.6
	2%	WB	15.8	19.2	23.2	26.2	28.4	29.1	29.5	29.3	28.7	26.4	21.1	17.3
		MCDB	19.5	23.6	29.6	36.1	38.1	35.6	33.5	32.8	32.3	30.9	26.2	21.1
	5%	WB	14.8	18.0	21.9	24.6	27.0	28.5	29.1	28.9	28.1	25.5	20.0	16.3
		MCDB	18.0	22.3	28.1	34.6	36.4	35.0	33.2	32.3	31.6	30.0	25.5	20.1
	10%	WB	13.9	16.9	20.7	23.2	25.8	27.9	28.7	28.5	27.5	24.6	18.9	15.4
		MCDB	16.8	20.7	26.8	33.2	35.4	34.1	32.7	31.8	31.0	29.2	24.2	19.2
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.2	12.7	15.2	13.6	10.5	6.4	6.2	8.0	12.4	13.6	11.4
		MCDBR	12.2	12.8	14.5	16.7	15.7	13.9	7.9	7.2	9.0	12.7	14.6	13.3
	5% WB	MCWBR	7.0	6.6	5.7	6.0	5.3	4.5	2.8	2.9	3.7	5.2	6.2	7.0
		MCWBR	10.7	11.8	13.5	14.9	13.8	9.9	7.0	6.6	7.9	10.9	13.1	12.1
Clear-Sky Solar Irradiance	taub	taub	0.474	0.477	0.495	0.565	0.668	0.747	0.758	0.643	0.567	0.636	0.582	0.515
		taud	1.892	1.935	1.928	1.804	1.634	1.538	1.544	1.771	1.888	1.636	1.693	1.802
	Ebn at Noon	Ebn at Noon	721	771	791	752	679	624	615	684	720	619	609	649
		Edh at Noon	164	172	185	217	259	283	281	221	190	230	200	173
All-Sky Solar Radiation	RadAvg	RadAvg	3.10	4.22	5.63	6.59	7.18	6.44	5.37	5.21	5.30	4.99	4.00	3.26
	RadStd	RadStd	0.31	0.39	0.30	0.28	0.24	0.49	0.42	0.34	0.35	0.26	0.19	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page